D02 · Supply Planning & Scheduling

01 · MRP / MPS mechanics & parameter optimization

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1 WHAT IT IS

MRP/MPS mechanics explode demand through the BOM against on-hand inventory and lead times to generate planned orders; parameter optimization tunes the levers that decide whether that plan is executable or nervous — lot-sizing policy, lead times, safety stock and safety time, and the time fences that freeze the near horizon against churn.

2 WHY IT MATTERS

Wagner & Whitin 1958 **(Wagner & Whitin, 1958 — peer-reviewed)** proved that the optimal lot-sizing policy under time-varying demand is neither lot-for-lot nor a fixed EOQ but a dynamic balance of setup against holding cost, period by period. MRP run on inherited fixed lot sizes ignores this and systematically mis-sizes orders. Worse, without time fences, MRP is structurally nervous: a small demand change at the top of the BOM cascades into wholesale re-timing of dependent orders every run. The result is a plan that changes every cycle — **planners stop trusting it and revert to manual firm orders**, and the expensive engine becomes a number generator no one executes.

3 HOW IT IS DONE

1. **Match the lot-sizing policy to the cost structure.** Lot-for-lot, fixed period, Wagner-Whitin, or POQ each suit a different setup-to-holding ratio. Inherited lot sizes are rarely optimal; segment items and assign the policy that balances their actual setup and carrying costs.
2. **Set time fences and a frozen horizon.** Freeze the near-term plan so the engine cannot re-time orders already in execution. The frozen and slushy/firm zones are what convert a nervous MRP into a stable one — without them, every run rewrites the schedule.
3. **Use firm planned orders to pin critical supply.** Lock orders the organization has committed to so MRP plans around them rather than through them. Firm planned orders are the planner's instrument for telling the engine what is no longer negotiable.
4. **Choose safety time vs safety stock by the source of variability.** Safety stock buffers quantity uncertainty; safety time buffers timing uncertainty. Applying the wrong one — stock for a lead-time problem — over-inventories without fixing the actual exposure.
5. **Validate planned lead times against actuals.** Planned lead times decay as the network changes; an MRP run on stale lead times schedules releases at the wrong time. Tie this to master-data governance — the engine inherits the quality of its parameters.
6. **Measure system nervousness explicitly.** Track how much the plan changes run-to-run inside the firm horizon. Rising churn is the leading indicator that planners are about to abandon the system; nervousness is a measurable parameter problem, not an inherent flaw of MRP.

WORKED EXAMPLE — ILLUSTRATIVE

A components plant ran MRP with go-live lot sizes and no time fences — **baseline: 34% of firm-horizon orders re-timed every weekly run, planners maintaining a parallel manual schedule, expedite-heavy execution**. It re-optimized lot-sizing by cost segment, set a four-week frozen horizon, and used firm planned orders on critical items. Results: **firm-horizon order churn 34% -> 8%, the parallel manual schedule retired, and planner trust in the system restored** — the math did not change, the parameters and the fences did.

IN PRACTICE

Configure and respect time fences

Define frozen, firm, and free zones aligned to the actual lead times of the supply chain, and hold the discipline of not replanning inside the frozen zone. The fence is worthless if planners override it routinely; stability comes from the discipline as much as the parameter.

Segment parameters, don't globalize them

Lot-sizing policy, safety stock, and fences should vary by item segment — high-volume stable items differ from intermittent or long-lead ones. A single global parameter set is a compromise that fits no segment well; segment-specific tuning is where the inventory and stability gains live.

Monitor a nervousness metric

Track run-to-run plan change inside the firm horizon as a standing KPI. A rising number is an early warning that the parameters are drifting or the fences are too loose — and a leading indicator of planners abandoning the system for manual control.

Tie parameters to master-data governance

Lead times and lot sizes are master data; their accuracy is governed by the same discipline that governs all planning data. Periodic validation against actuals keeps the MRP engine fed with current parameters rather than the values frozen at implementation.

WHAT YOU WOULD USE

MRP / MPS / SUPPLY PLANNING ENGINES

SAP (ECC/S4 MRP, IBP Response) / Oracle — enterprise tier-1 — Full MRP/MPS with configurable lot-sizing, time fences, and firm planned orders integrated to ERP execution. **(SAP, Oracle — vendor)** The choice when the supply plan must drive procurement and production execution directly from one system of record.

Kinaxis / o9 / Blue Yonder — advanced supply planning — APS engines that add constraint awareness, fast re-planning, and stronger parameter analytics on top of MRP logic. **(Kinaxis, o9, Blue Yonder — vendor)** A strong fit where nervousness and constraint handling are the binding problems.

Parameter-optimization / inventory-tuning specialists — Dedicated tools that optimize lot sizes, safety stock, and lead-time parameters across the item base. **(specialized vendors — vendor)** The vertical option when the core engine is fine but its parameters have never been systematically tuned.

For operations that don't need a platform — A disciplined lot-sizing review by cost segment, configured time fences in the existing ERP MRP, and a nervousness KPI tracked in a spreadsheet. Most MRP pain is parameter and discipline, not engine — the levers already exist in nearly every ERP.

4 SO WHAT

Operations that optimize lot sizes and freeze the near horizon get a plan stable enough to execute; operations that run go-live parameters get a nervous engine and a shadow manual schedule. The most common MRP failure is not the algorithm — it is parameters set once and time fences never configured. System nervousness is a measurable, fixable parameter problem, and the planner override rate is the bill for ignoring it.

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